

Numerical modelling of turbulent iron combustion

PROEFSCHRIFT

ter verkrijging van de graad van doctor aan de Technische Universiteit
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commissie aangewezen door het College voor Promoties, in het openbaar te
verdedigen op dag XX maand jaar om YY:ZZ

door

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geboren te Coimbatore, India

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adviseur: naam (bedrijf)

Het onderzoek dat in dit proefschrift wordt beschreven is uitgevoerd in overeenstemming met de TU/e Gedragscode Wetenschapsbeoefening.

“Very deep quote”

Some old guy, long ago

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Summary

Samenvatting

Contents

Summary	v
Samenvatting	vii
1 Introduction	1
1.1 The iron power cycle	3
1.2 Kinetics of micron-sized iron particle combustion	3
1.3 Turbulent iron powder flames	3
1.3.1 Multiphase phenomena involving turbulence	3
1.3.2 Large-scale industrial combustors	3
1.4 Goal and outline of the dissertation	3
2 Numerical framework for simulation of turbulent iron combustion	5
2.1 Introduction.	7
2.2 Eulerian-Lagrangian framework	7
2.2.1 Gas-phase model	7
2.2.2 Particle model.	7
2.2.3 Two-way coupling	7
2.3 Implementation.	7
2.3.1 Direct numerical simulation in NTMIX-CHEMKIN.	7
2.3.2 Large eddy simulation in <i>OpenFOAM</i>	7
2.4 Validation	7
2.4.1 Limitations	7
2.5 Summary	7
3 Theoretical analysis of timescales in turbulent iron combustion	8
3.1 Introduction.	10
3.1.1 Motivations and goals	10
3.2 Timescales of iron particle clustering	10
3.2.1 Cluster formation	10
3.2.2 Cluster evolution	10
3.3 Timescales of iron particle combustion	10
3.3.1 Ignition	10
3.3.2 Oxidation	10
3.4 Comparison of timescales	10
3.4.1 Regime map of turbulent iron combustion.	10
3.4.2 Example cases.	10
3.5 Summary	10
4 Direct numerical simulations of forced HIT field	11
4.1 Introduction.	13
4.1.1 Motivation and goals	13

4.2	Simulation setup	13
4.3	Analysis setup.	13
4.4	Results and discussion	13
4.4.1	Visualization	13
4.4.2	Trends of clustering timescales and magnitudes.	13
4.4.3	Effects of combustion on particle clustering	13
4.4.4	Global temperature evolutions	13
4.4.5	Burnout times at various Re and ϕ	13
4.4.6	Correlating spatial and combustion properties	13
4.5	Summary	13
5	Direct numerical simulations of a reacting mixing layer	14
5.1	Introduction.	16
5.1.1	Motivation and goals	16
5.2	Simulation setup	16
5.3	Analysis setup.	16
5.4	Results and discussion	16
5.4.1	Visualization	16
5.4.2	Perturbations of the mixing layer	16
5.4.3	Analysis of laminar-like combustion mode.	16
5.4.4	Analysis of turbulent-like combustion mode.	16
5.5	Summary	16
6	Large eddy simulations of jet-in-hot-coflow burner	17
6.1	Introduction.	19
6.1.1	Motivation and goals	19
6.2	Simulation setup	19
6.3	Analysis setup.	19
6.4	Results and discussion	19
6.4.1	Visualization	19
6.4.2	Flame structure at criticality	19
6.4.3	Overall ignition characteristics	19
6.4.4	What causes partial oxidation?	19
6.4.5	Assessment of assumptions.	19
6.5	Summary	19
7	Direct numerical simulations of ignition in spherical suspensions	20
7.1	Introduction.	22
7.1.1	Motivation and goals	22
7.2	Simulation setup	22
7.3	Analysis setup.	22
7.4	Results and discussion	22
7.4.1	Visualization	22
7.4.2	Ignition delay at various initial temperatures	22
7.4.3	Where does ignition occur?	22
7.5	Summary	22

8	Conclusions and Outlook	23
8.1	Conclusions	25
8.2	Outlook	25
A	Appendices	26
A.1	Supplementary of Chapter 2	27
A.2	Supplementary of Chapter 3	27
A.3	Supplementary of Chapter 4	27
A.4	Supplementary of Chapter 5	27
A.5	Supplementary of Chapter 6	27
A.6	Supplementary of Chapter 7	27
	References	29
	Publications & conference contributions	31
	Acknowledgements	33
	About the author	35

1

Introduction

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Abstract

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1.1. The iron power cycle

...

1.2. Kinetics of micron-sized iron particle combustion

...

1.3. Turbulent iron powder flames

...

1.3.1. Multiphase phenomena involving turbulence

1.3.2. Large-scale industrial combustors

1.4. Goal and outline of the dissertation

This thesis is organized as follows. In Chapter 2 ...

In Chapter 3, ...

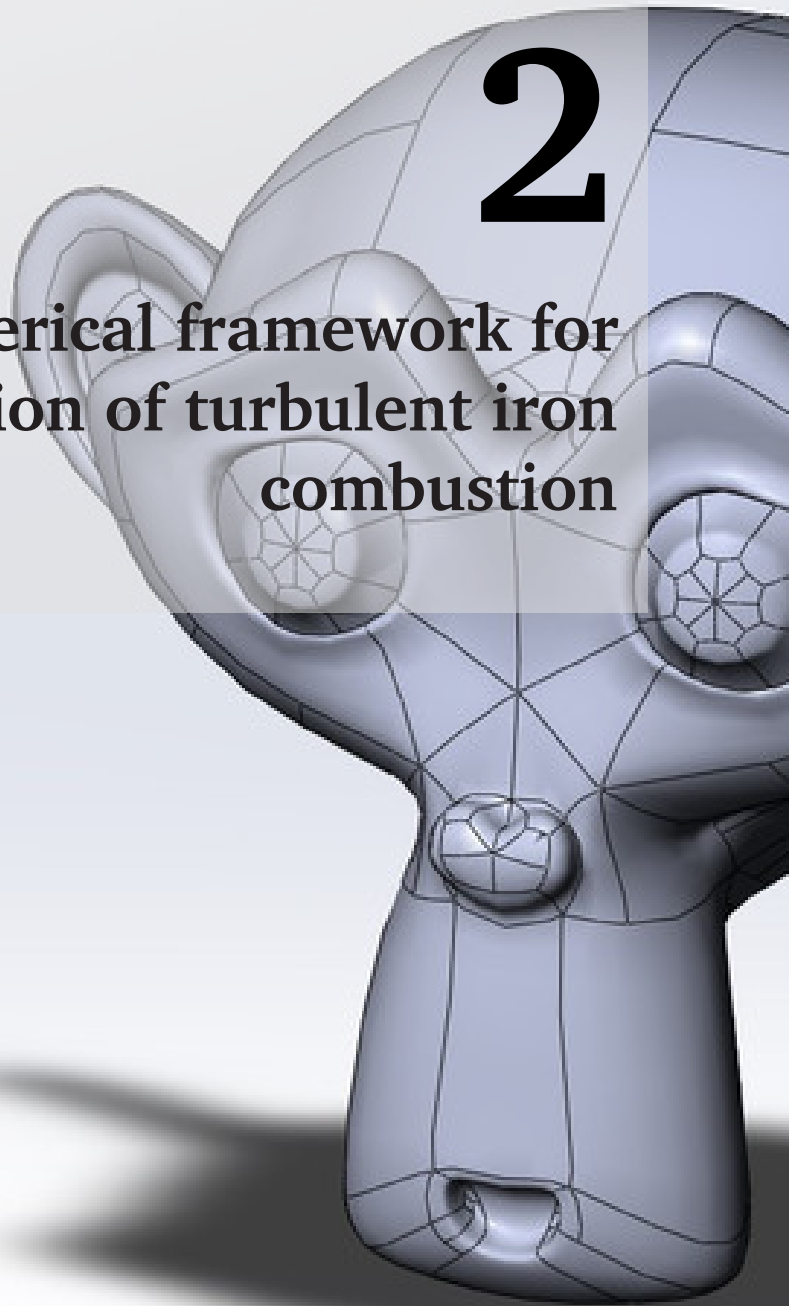
In Chapter 4, ...

In Chapter 5, ...

Chapter 8 summarizes the work in this thesis. Recommendations are provided for further research/investigation on ...

2

**Numerical framework for
simulation of turbulent iron
combustion**



Abstract

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This work used the Dutch national e-infrastructure with the support of the SURF Cooperative using grant no. EINF-8301.

2.1. Introduction

Citing something to show that it works [1]. Reference to Chapter 1. More info can be found in Appendix A.1.

2.2. Eulerian-Lagrangian framework

2.2.1. Gas-phase model

2.2.2. Particle model

2.2.3. Two-way coupling

2.3. Implementation

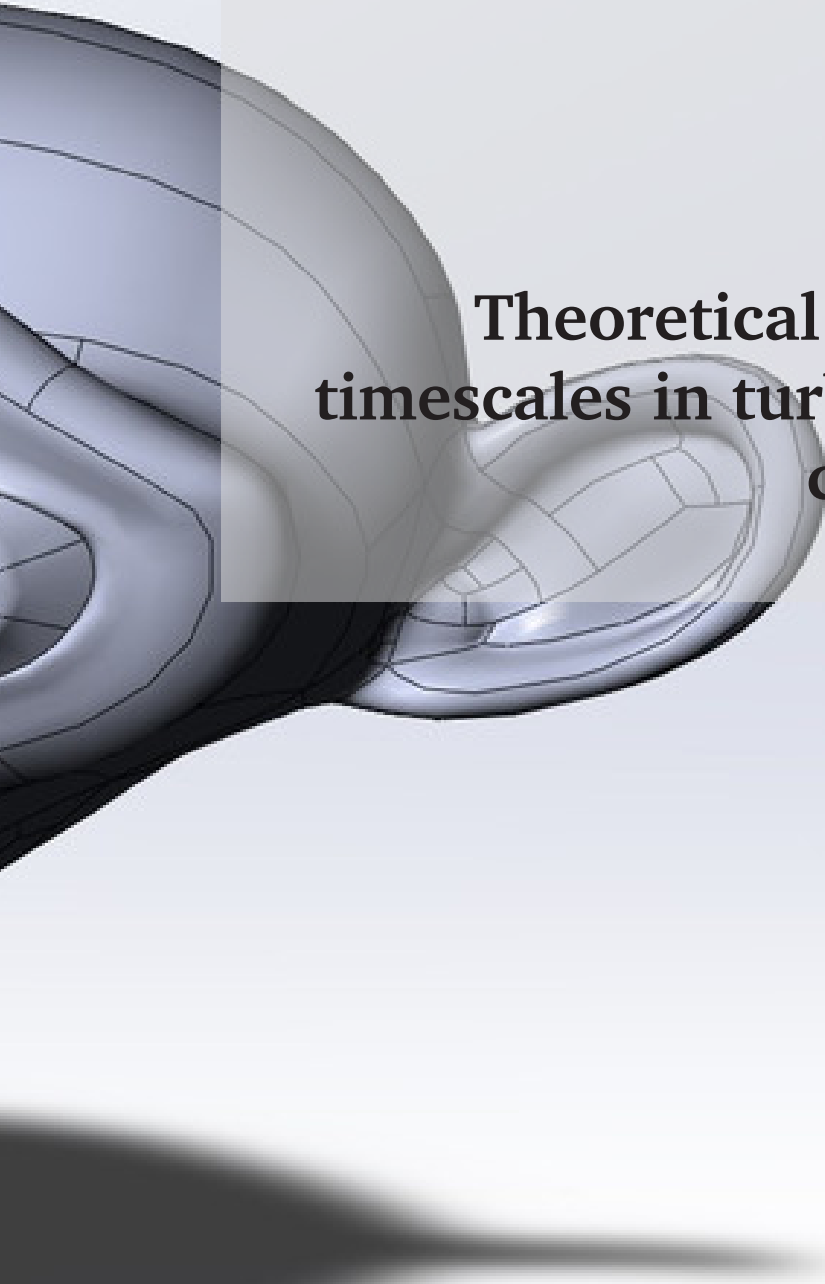
2.3.1. Direct numerical simulation in NTMIX-CHEMKIN

2.3.2. Large eddy simulation in *OpenFOAM*

2.4. Validation

2.4.1. Limitations

2.5. Summary



3

Theoretical analysis of timescales in turbulent iron combustion

*Parts of this chapter are published in:
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Abstract

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3.1. Introduction

...

3.1.1. Motivations and goals

3.2. Timescales of iron particle clustering

3.2.1. Cluster formation

3.2.2. Cluster evolution

3.3. Timescales of iron particle combustion

3.3.1. Ignition

3.3.2. Oxidation

3.4. Comparison of timescales

3.4.1. Regime map of turbulent iron combustion

3.4.2. Example cases

3.5. Summary

4

Direct numerical simulations of forced HIT field

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Abstract

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4.1. Introduction

4.1.1. Motivation and goals

4.2. Simulation setup

4.3. Analysis setup

4.4. Results and discussion

4.4.1. Visualization

4.4.2. Trends of clustering timescales and magnitudes

4.4.3. Effects of combustion on particle clustering

4.4.4. Global temperature evolutions

4.4.5. Burnout times at various Re and ϕ

4.4.6. Correlating spatial and combustion properties

Underlying trends in τ_b^* vs. V_{norm}

Time-averaging spatial properties

Significance of macroscale structures

4.5. Summary

5

Direct numerical simulations of a reacting mixing layer

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Abstract

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5.1. Introduction

5.1.1. Motivation and goals

5.2. Simulation setup

5.3. Analysis setup

5.4. Results and discussion

5.4.1. Visualization

5.4.2. Perturbations of the mixing layer

5.4.3. Analysis of laminar-like combustion mode

5.4.4. Analysis of turbulent-like combustion mode

5.5. Summary

6

Large eddy simulations of jet-in-hot-coflow burner

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Abstract

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6.1. Introduction

6.1.1. Motivation and goals

6.2. Simulation setup

6.3. Analysis setup

6.4. Results and discussion

6.4.1. Visualization

6.4.2. Flame structure at criticality

6.4.3. Overall ignition characteristics

6.4.4. What causes partial oxidation?

6.4.5. Assessment of assumptions

Initial particle clustering

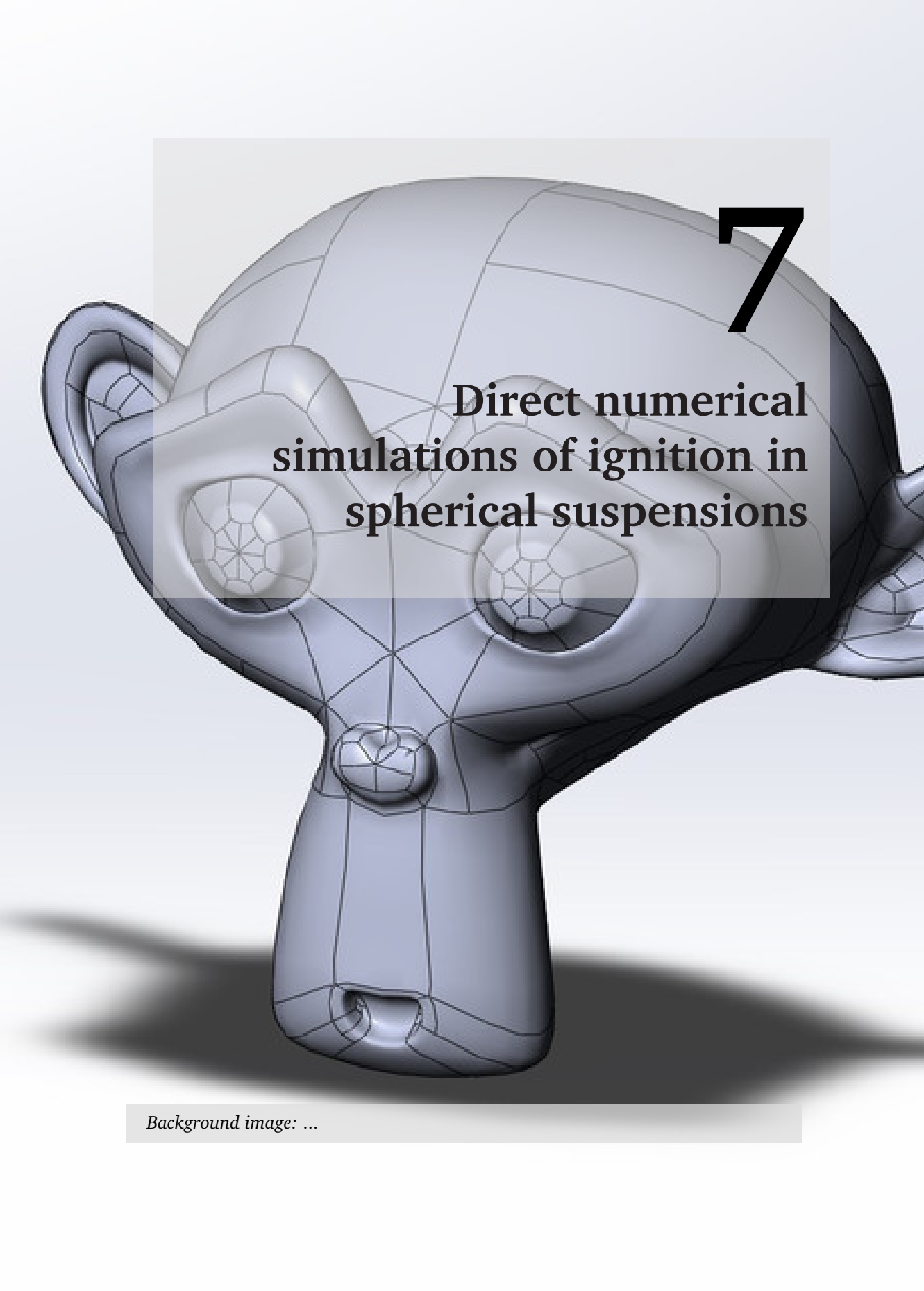
Particle injection temperature

Injection velocity and temperature profiles

Radial size preference at injection

Material characteristics

6.5. Summary



7

**Direct numerical
simulations of ignition in
spherical suspensions**

Background image: ...

Abstract

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7.1. Introduction

7.1.1. Motivation and goals

7.2. Simulation setup

7.3. Analysis setup

7.4. Results and discussion

7.4.1. Visualization

7.4.2. Ignition delay at various initial temperatures

7.4.3. Where does ignition occur?

Auto-ignition vs. thermal wave propagation

Localized heating through clustering

7.5. Summary

8

Conclusions and Outlook

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Abstract

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8.1. Conclusions

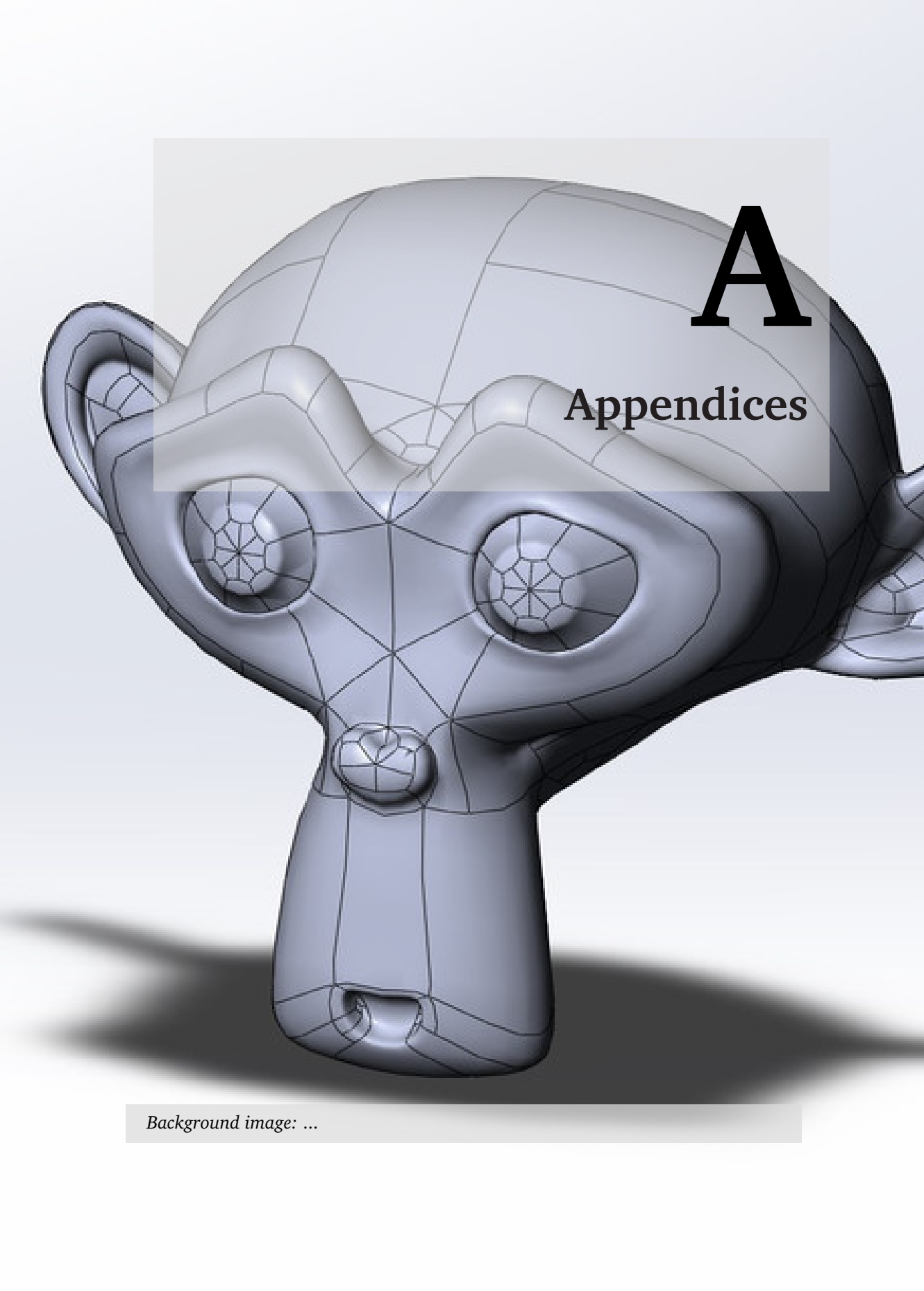
The findings from each study are summarized in the individual chapters. Here, the overall findings/insights are summarized in the following:

- ...
- ...
- ...
- ...

8.2. Outlook

However, many new research questions also arise, which were not investigated due to time and recourse restrictions. These can be summarized as:

- ...
- ...
- ...
- ...



A

Appendices

Background image: ...

A.1. Supplementary of Chapter 2

This appendix discusses in greater detail the derivation and implementation of the phase fraction equation and pressure correction step in the solver *compressibleInterIsoFoam* in OpenVFOAM[®] version 24.06.

A.2. Supplementary of Chapter 3

A.3. Supplementary of Chapter 4

A.4. Supplementary of Chapter 5

A.5. Supplementary of Chapter 6

A.6. Supplementary of Chapter 7

References

- [1] M. W. Baltussen, J. A. M. Kuipers, and N. G. Deen. “A critical comparison of surface tension models for the volume of fluid method”. In: *Chemical Engineering Science* 109 (2014), pp. 65–74. DOI: 10.1016/j.ces.2013.12.045.

Publications & conference contributions

Publications

1. D.P.L. Thuy, N.G. Deen, J.J.C. Remmers, G. Finotello. Volume-of-Fluid simulations of multiphase flows with high surface tension and curvature: a tensile force approach with pressure jump correction. *Journal*, 123 (2025) 4567890.

Conference contributions

1. **D.P.L. Thuy**, J.J.C. Remmers, N.G. Deen, G. Finotello, Large eddy simulations of primary breakup in metal melt gas atomization (presentation, video), *Burgers Symposium*, Lunteren, The Netherlands (2023).
2. **D.P.L. Thuy**, J.J.C. Remmers, N.G. Deen, G. Finotello, Large eddy simulations of primary breakup in metal melt gas atomization (presentation), *11th International Conference on Multiphase Flow*, Kobe, Japan (2023).
3. **D.P.L. Thuy**, J.J.C. Remmers, N.G. Deen, G. Finotello, Large eddy simulations of primary breakup in metal melt gas atomization (presentation), *International Conference on Numerical Methods in Multiphase Flows 4*, Venice, Italy (2022).
4. **D.P.L. Thuy**, J.J.C. Remmers, N.G. Deen, G. Finotello, Primary breakup modeling in metal melt gas atomization (presentation), *The 17th OpenFOAM Workshop*, Cambridge, United Kingdom (2022).

Acknowledgements

About the author

Dennis Thuy was born on the 5th of December 1997 in Sint Jansteen, the Netherlands.

